



Koneru Lakshmaiah Education Foundation

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FEDF CO-3

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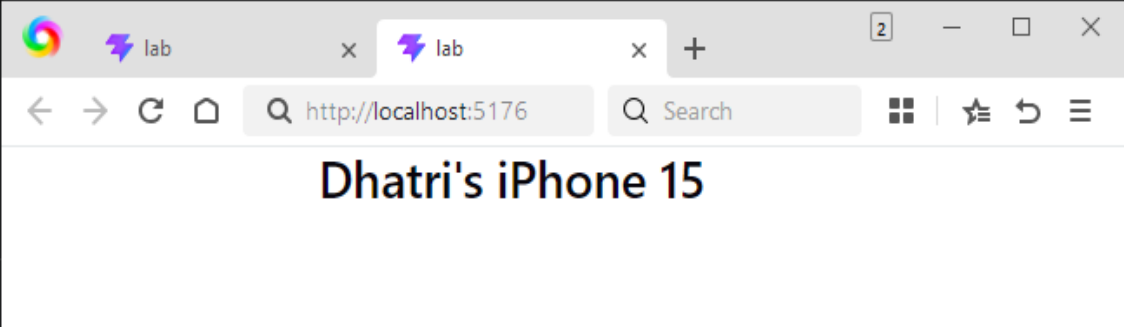
Roll No: 2520040012

Guide: K. Anusha

Topic: React Programs Output Screenshots

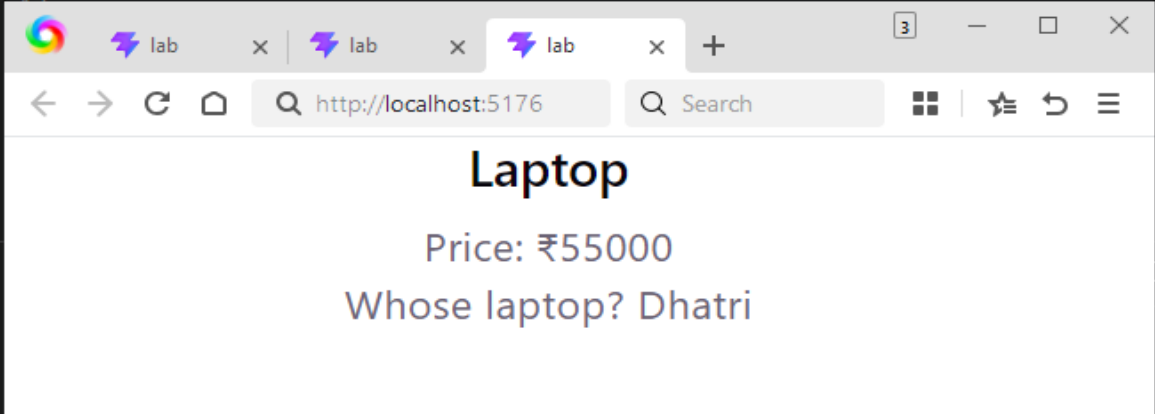
1) Component as Deterministic UI Function

```
lab > src > App.jsx > default
1  function Product(props) {
2    return <h2>{props.name}</h2>;
3  }
4
5  function App() {
6    return <Product name="Dhatri's iPhone 15" />;
7  }
8  export default App;
```



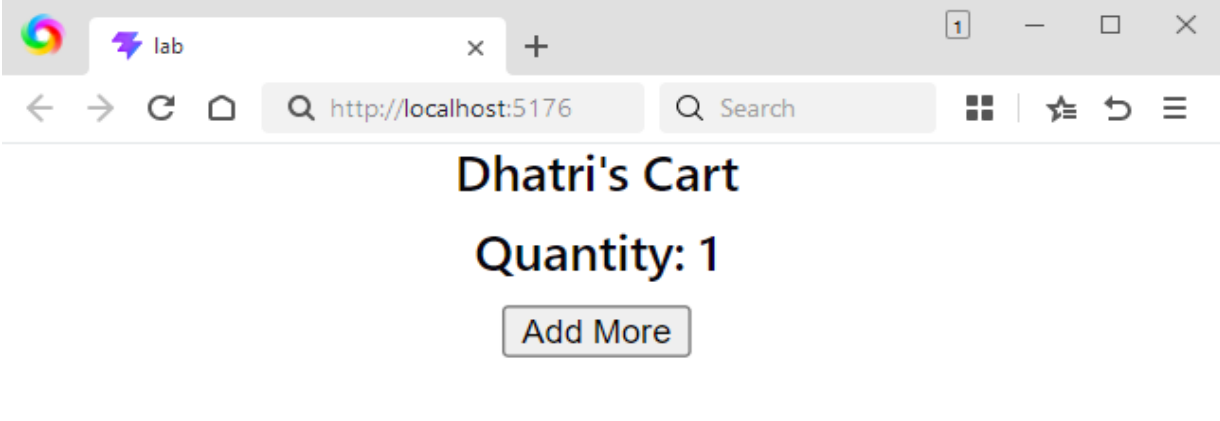
2) Props as Configuration Contract

```
lab > src > App.jsx > App
1  function Product(props) {
2    // ...
9  }
10
11 function App() {
12   return (
13     <Product name="Laptop" price="55000" owner="Dhatri" />
14   );
15 }
16 export default App;
```



3) State as Dynamic Runtime

```
3  function Cart() {
4    // ...
5
6    return (
7      <div>
8        <h2>Dhatri's Cart</h2>
9        <h2>Quantity: {quantity}</h2>
10
11        <button onClick={() => setQuantity(quantity + 1)}>
12          Add More
13        </button>
14      </div>
15    );
16  }
17  export default Cart;
```

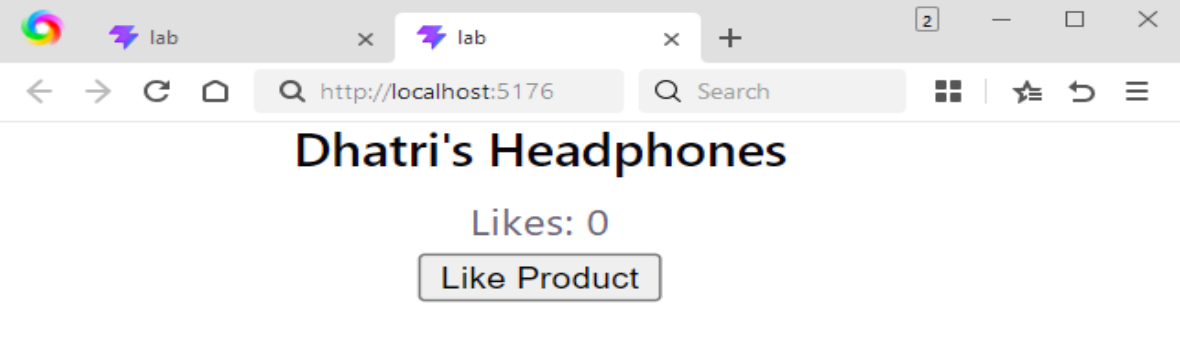


4) Props vs State

Example: Product with Dynamic Likes

```
lab > src > App.jsx > Product
3  function Product(props) {
6    return (
7      <div>
8        <h2>Dhatri's {props.name}</h2>
9
10       <p>Likes: {likes}</p>
11
12       <button onClick={() => setLikes(likes + 1)}>
13         Like Product

```



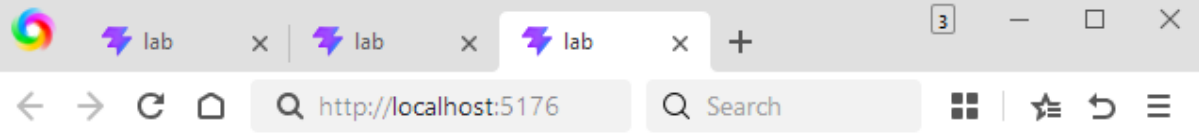
The screenshot shows a web browser window with the URL `http://localhost:5176`. The page displays the text "Dhatri's Headphones" in a large, bold font. Below it, the text "Likes: 0" is shown. At the bottom, there is a button labeled "Like Product".

5) Side Effects using useEffect

Example: Fetch Product Data

```
lab > src > App.jsx > ...
1  import { useEffect } from "react";
2
3  function ProductPage() {
4
5    useEffect(() => {
6      console.log("Fetching Product Details...");
7    }, []);
8
9    return <h1>Product Details for Dhatri</h1>;

```



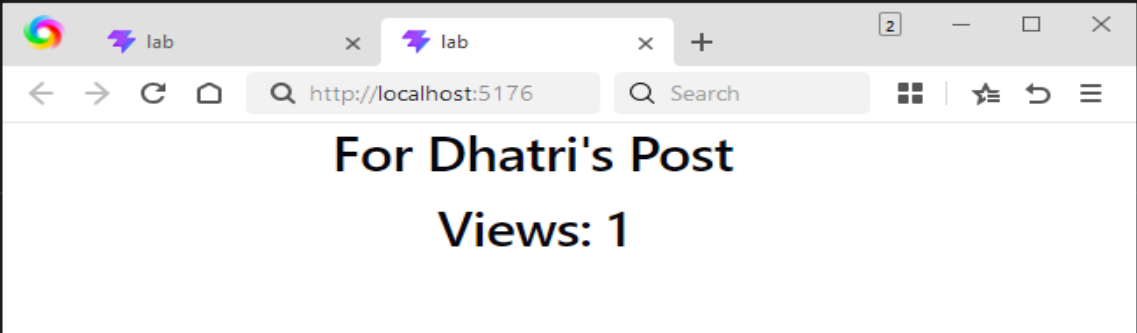
The screenshot shows a web browser window with the URL `http://localhost:5176`. The page displays the text "Product Details for Dhatri" in a large, bold font.

Product Details for Dhatri

6) Hooks as Lifecycle Abstraction

Example: Product View Counter

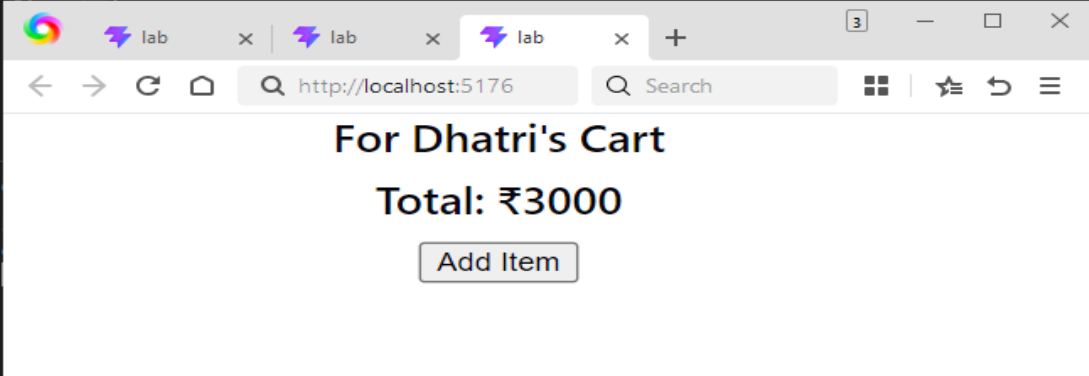
```
lab > src > App.jsx > Product
3  function Product() {
10
11  return (
12    <div>
13      <h2>For Dhatri's Post</h2>
14      <h2>Views: {views}</h2>
15    </div>
16  );
17 }
18 export default Product;
```



7) Reconciliation (Re-rendering)

Example: Cart Total Update

```
lab > src > App.jsx > Cart
3  function Cart() {
6
7  return (
8    <div>
9      <h2>For Dhatri's Cart</h2>
10     <h2>Total: ₹{total}</h2>
11
12     <button onClick={() => setTotal(total + 500)}>
13       Add Item
14     </button>
15   </div>
16 )
```

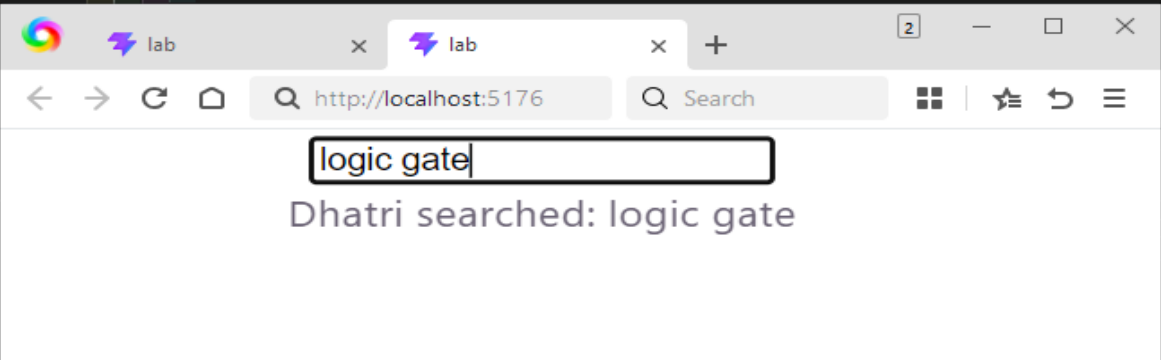


8) Controlled Component

Example: Search Bar

```
lab > src > App.jsx > Search
3  function Search() {
4
5      const [text, setText] = useState("");
6
7      return (
8          <div>
9              <input
10                 type="text"
11                 value={text}
12                 onChange={(e) => setText(e.target.value)}
13                 placeholder="Search Product"

```

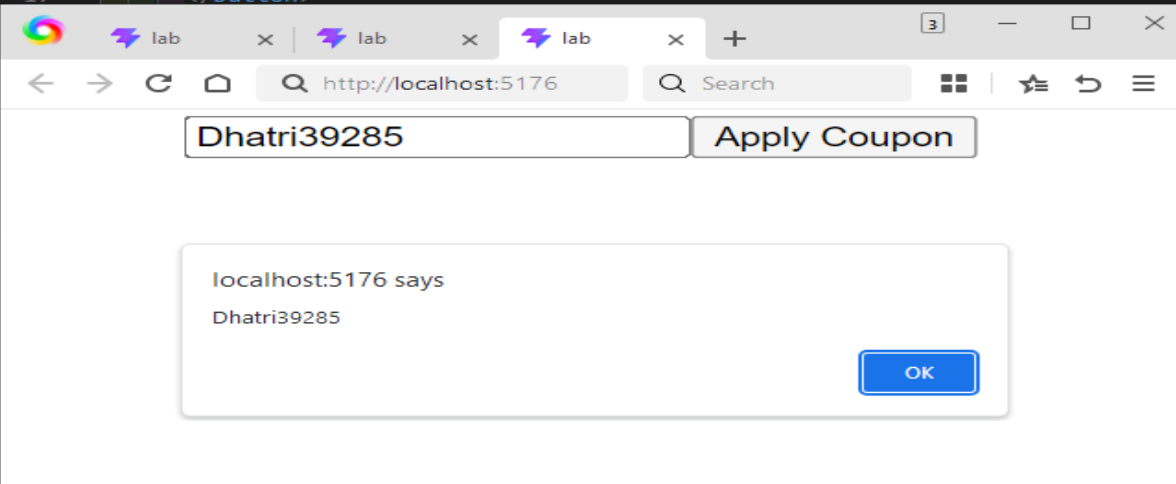


9) Uncontrolled Component

Example: Coupon Code Input

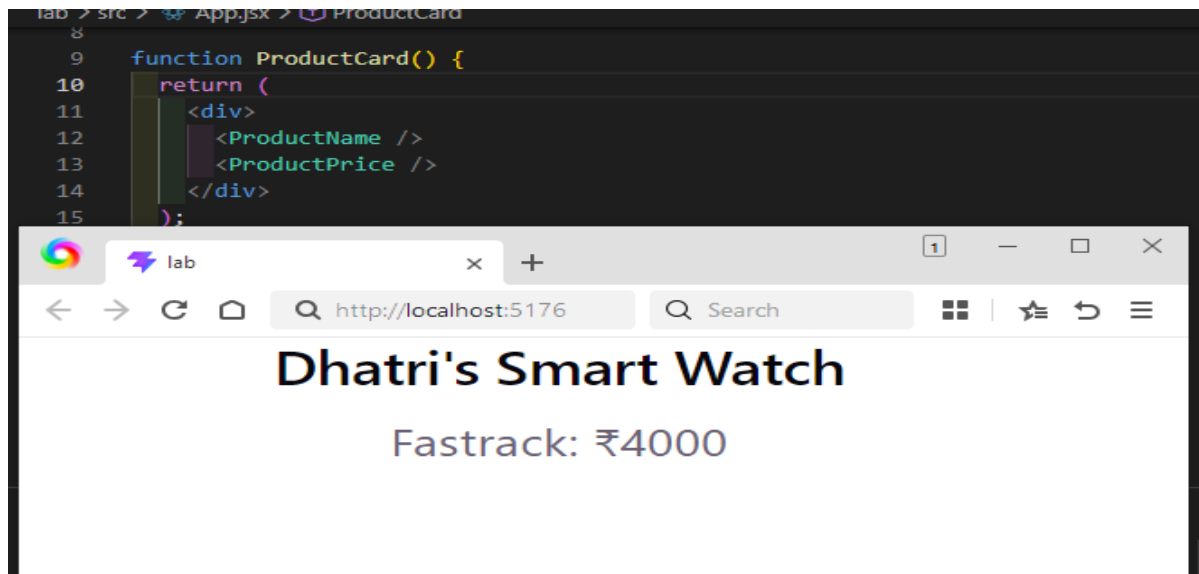
```
3  function Coupon() {
11     return (
12         <div>
13             <input type="text" ref={inputRef} />
14
15             <button onClick={applyCoupon}>
16                 Apply Coupon
17             </button>

```



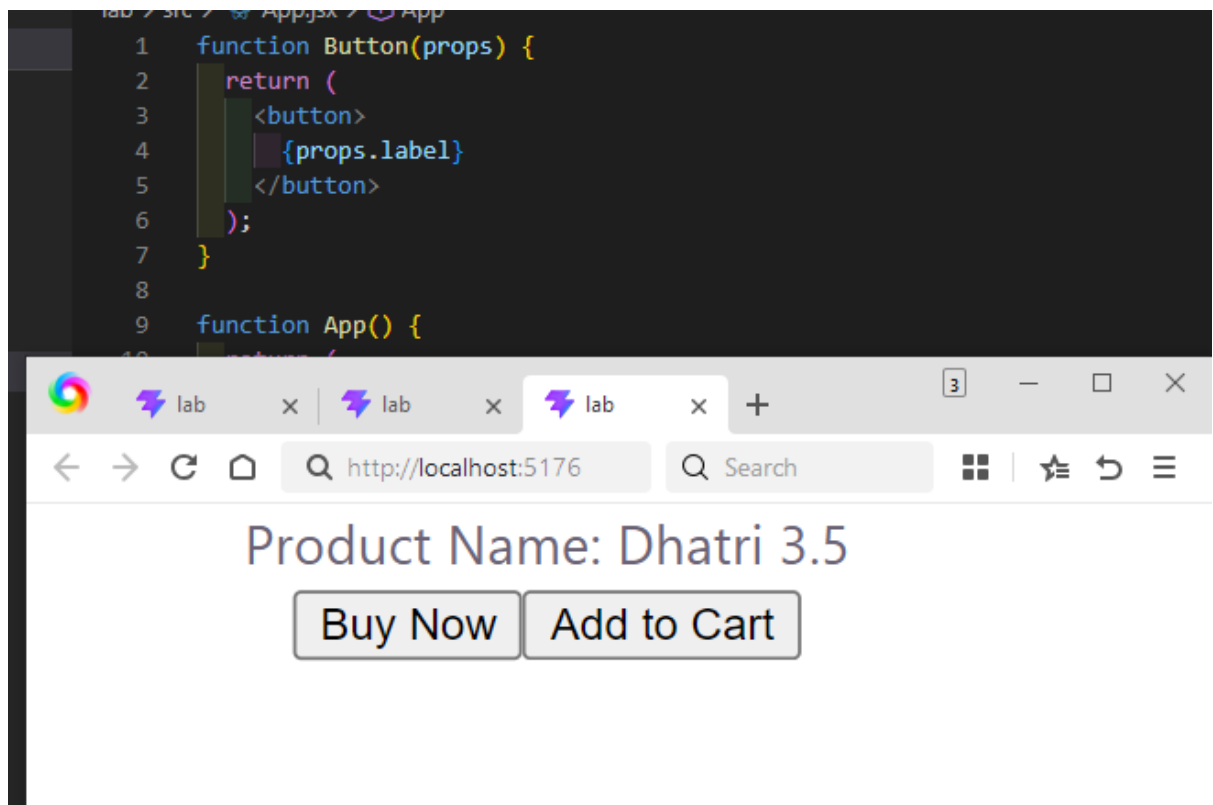
10) Component Composition

Example: Product Card using Smaller Components



11) Reusable Components

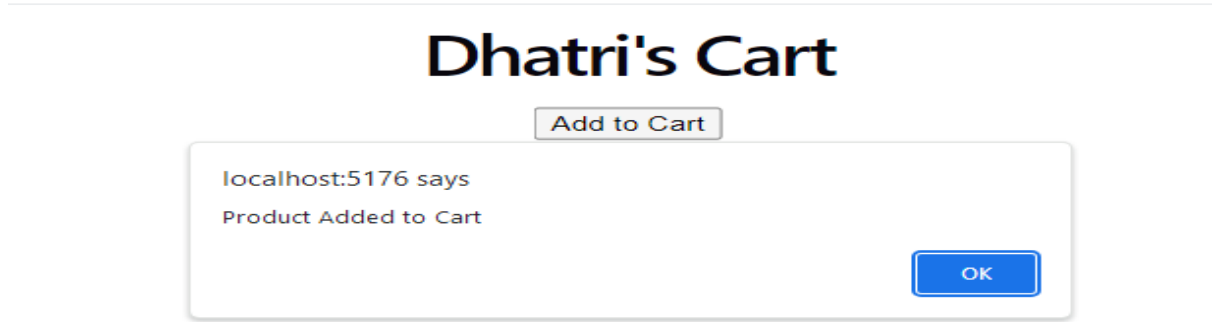
Example: Reusable Button



12) Event Handler Example

Example: Add to Cart

```
1 function Product() {  
2   return (  
3     <div>  
4       <h1>Dhatri's Cart</h1>  
5       <button onClick={addToCart}>  
6         Add to Cart  
7       </button>  
8     </div>  
9   );  
10 }  
11  
12  
13  
14  
15
```



13) Children Props Example

Example: Product Card Layout

```
lab > src > App.jsx > default  
1 function Card(props) {  
2   return (  
3     {props.children}  
4   </div>  
5   );  
6 }  
7  
8  
9 function App() {  
10  return (  
11    <Card>  
12      Dhatri's Dress Cost  
13      ₹2000  
14    </Card>  
15  )  
16 }
```

